MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL MX-9872(V)7/UA

This specification sheet is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the MX-9872(V)7/UA Scanner shall consist of this document and MIL-S-49166(ER).

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-9872(V)7/UA and augments MIL-S-49166. The MX-9872(V)7/UA is used on the M-65 C-Nite Thermal Imaging System.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49166 to describe the MX-9872(V)7/UA Mechanical Scanner. All the MIL-S-49166 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49166 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect or date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-49166(CR) - Scanner, Mechanical, MX-9872(V)/UA, General Specifications for

DRAWINGS

USA Electronics Command Drawings

80063 SM-D-807690 - Mechanical Scanner Assembly, MX-9872(V)/UA
80063 SM-D-657598 - Scan-Interlace PL-1403A/UA
80063 SM-D-773950 - Scan-Interlace PL-1403/UA

2.2 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Imrpovement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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3. REQUIREMENTS

3.1 Description. The MX-9872(V)7/UA Scanner, herein known as the MX-9872(V)7/UA, shall consist of the MIL-S-49166, MX-9872(V)/UA Scanner adjusted for a nominal 30.0 Hz and 8.0 degree scan angle operation. The MX-10717(V)7/U Scanner can be operated with either Scan-Interlace Assembly PL-1403/UA (SM-D-773950) on PL-1403A/UA (SM-D-657598). These two assemblies will be referred to as the Scan-Interlace Assembly throughout this specification sheet.

3.2 Construction. The MX-9872(V)7/UA shall be constructed in accordance with drawing 80063 SM-D-807690-4.

3.2.1 Weight. Weight of the MX-9872(V)7/UA shall be 1.90 pounds maximum.

3.6 Performance characteristics. The Characteristics specified herein shall apply as constructed (3.2). Operation shall be at a nominal 30.0 Hz, 2:1 rear interlace mode while using the Scan-Interlace Assembly.

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Not applicable.

3.6.1.2 Specific scan frequency (30.0 HZ). Scan frequency of the MX-9872(V)7/UA shall be adjusted to 30.0 + 0.5 Hz. Scan frequency drift shall not be greater than ± 1.5 Hz from the nominal frequency at +73 ± 9 degrees F when exposed to high and low temperature environmental conditions (see MIL-S-49166).

3.6.1.3 Synchronous scan frequency. Not applicable.

3.6.1.4 Scan angle. The scan angle shall be 4.0 ± 0.1 degrees in each direction (8.0 ± 0.2 degrees total) from a true 45 degree position with respect to the optical axis (DATUM -A- in figure 1).

3.6.2 Interlace. The MX-9872(V)7/U shall operate in a 2:1 rear interlace mode. The switching time shown in figure 1 of MIL-S-49166 shall not be greater than 3.5 milliseconds (ms) and the average amplitude shall be 1.0 ± 0.2 Vp-p.

3.6.2.1 Interlace angle. The interlace solenoid stops and Scan-Interlace Assembly potentiometers shall be used to determine an object space interlace angle of 155 ± 15 arc seconds when used with the Scan-Interlace Assembly.

3.6.2.2 Elevation boresight. The scan mirror shall not deviate by more than ± 16 arc minutes in elevation from the scanner (DATUM -A- mounting surface). See figure 2.

3.6.2.3 Interlace transducer D.C. offset. The interlace position transducer shall have a D.C. offset of 0.0 ± 0.2 Vdc.

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3.6.3 Scan direction. The MX-9872(V)7/UA shall be in accordance with the waveform shapes and phase relationships in figure 3 while operating in the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the MX-9872(V)7/UA connected to the Scan-Interlace Assembly shall not exceed 2.2 watts total.

3.6.5 Scan efficiency. The MX-9872(V)7/UA scan efficiency shall be a minimum of 75 percent.

3.6.6 Scan jitter in azimuth. Horizontal jitter shall not exceed 62.0 microseconds.

3.7 Environmental conditions. See MIL-S-49166 except as follows:

3.7.4 Shock. See MIL-S-49166 and 3.7.4.1.

3.7.4.1 Shock and scan frequency. Scan frequency (3.6.1.2) shall return to normal within 0.2 seconds after the MX-9872(V)7/UA is subjected to the operating shock environment.

3.7.5 Vibration. The MX-9872(V)7/UA shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 5 for the X axis and as shown in figure 3 of MIL-S-49166 for the Y and Z axes. During specified vibration, the average interlace waveform amplitude shown in MIL-S-49166 figure 1 shall not deviate from the average amplitude without vibration by more than ± 50 percent.

4. QUALITY ASSURANCE PROVISION

4.3 First article inspection. See MIL-S-49166.

4.3.2 Test. In addition to the requirements of MIL-S-49166, Table II, the requirements of 3.6.2.1, 3.6.2.2 and 3.6.2.3 shall also be met. Failure of any tests shall constitute failure of first article inspection.

4.5 Quality conformance inspection.

4.5.2 Tests.

4.5.2.1 Group A inspection. In addition to the requirements of MIL-S-49166, Table III, the requirements of paragraphs 3.6.2.1, 3.6.2.2 and 3.6.2.3 and test paragraphs 4.6.2.1, 4.6.2.2 and 4.6.2.3 shall also be met. Failure of any test shall be cause for rejection of the unit.

4.6 Test methods. Tests shall be performed utilizing the Scan-Interlace Assembly with the MX-9872(V)7/UA.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. Not applicable.

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4.6.1.2 Specific scan frequency (30Hz). The MX-9872(V)7/UA shall be connected to the Scan-Interlace Assembly and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) tachometer (Scan-Interlace Assembly) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace Assembly shall be adjusted to obtain the specified frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan fequency. Not applicable.

4.6.1.4 Scan angle.

4.6.1.4.1 Test apparatus. The test apparatus shall consist of a holding fixture required for securing the MX-9872(V)7/UA, a projection system capable of projecting a spot or marker into the scanner along the optical axis (figure 1), and an appropriate scale graduated in degrees measured about the scan mirror axis. The zero position of this scale shall be the position where the spot or marker is projected onto the scale when the mirror is at a true 45 degrees with respect to the optical axis (DATUM -A- in figure 1).

4.6.1.4.2 Test conduction. The scanner shall be operated and the angular excursions of the projected spot, in each direction, shall be recorded.

4.6.1.4.3 Failure criteria. Failure to meet the requirement of 3.6.1.4 shall constitute failure of this test.

4.6.2 Interlace. With the scanner mounted in a suitable fixture, the output of the interlace position transducer shall be monitored with an oscilloscope. The waveform switching times (figure 1) of MIL-S-49166 and amplitude shall be measured to verify that the requirements of 3.6.2 are met. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.2.1 Interlace angle. Using an autocollimator and an auxiliary mirror to measure along the optical axis, the interlace solenoid stops and Scan-Interlace Assembly shall be adjusted to provide a dynamic object space interlace angle of 155 $\pm$ 15 arc seconds. Interlace angle shall be measured with the interlace operating. Failure to meet the requirements of 3.6.2.1 shall constitute failure of this test.

4.6.2.2 Elevation boresight. The scan mirror shall not deviate by more than $\pm$ 16 arc minutes in elevation from the optical axis (see figure 2). Elevation boresight shall be measured at the approximate center of the azimuth scan travel with interlace not operating. Failure to meet the requirements of 3.6.2.2 shall constitute failure of this test.

4.6.2.3 Interlace transducer D.C. offset. The D.C. offset of the interlace position transducer shall be measured at J3 of the Scan-Interlace Assembly with the interlace gimbal in the mechanical zero position. The D.C. offset shall meet the requirements of 3.6.2.3. Failure to meet the requirements of 3.6.2.3 shall constitute a failure of this test.

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4.6.3 Scan direction. With the MX-9872(V)7/UA operating in the 2:1 rear interlace mode, the outputs of the scan tachometer (J2) and interlace position transducer (J3) (Scan-Interlace Assembly) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationship shown in figure 3. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49166.

4.6.5 Scan efficiency. The MX-9872(V)7/UA shall be operated with the Scan-Interlace Assembly and measurements taken as follows: The period of the scan tachometer is measured by an oscilloscope. The active time between two consecutive scans shall be determined by measuring the time between the 10 percent and the 90 percent points of the scan tachometer waveform (J2). Scan efficiency shall be calculated by dividing the sum of the two consecutive scans by the scan tachometer period and multiplying by 100 percent. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Scan jitter in azimuth. The MX-9872(V)7/UA shall be tested as specified in MIL-S-49166 with additional circuitry used as shown in figure 4. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.7 Environmental tests.

4.7.4 Shock. See MIL-S-49166 and 4.7.4.1

4.7.4.1 Shock and scan frequency. Failure to meet the requirements of 3.7.4.1 during the operating test shall constitute failure of this test.

4.6.5 Vibration. See MIL-S-49166 and figure 5.

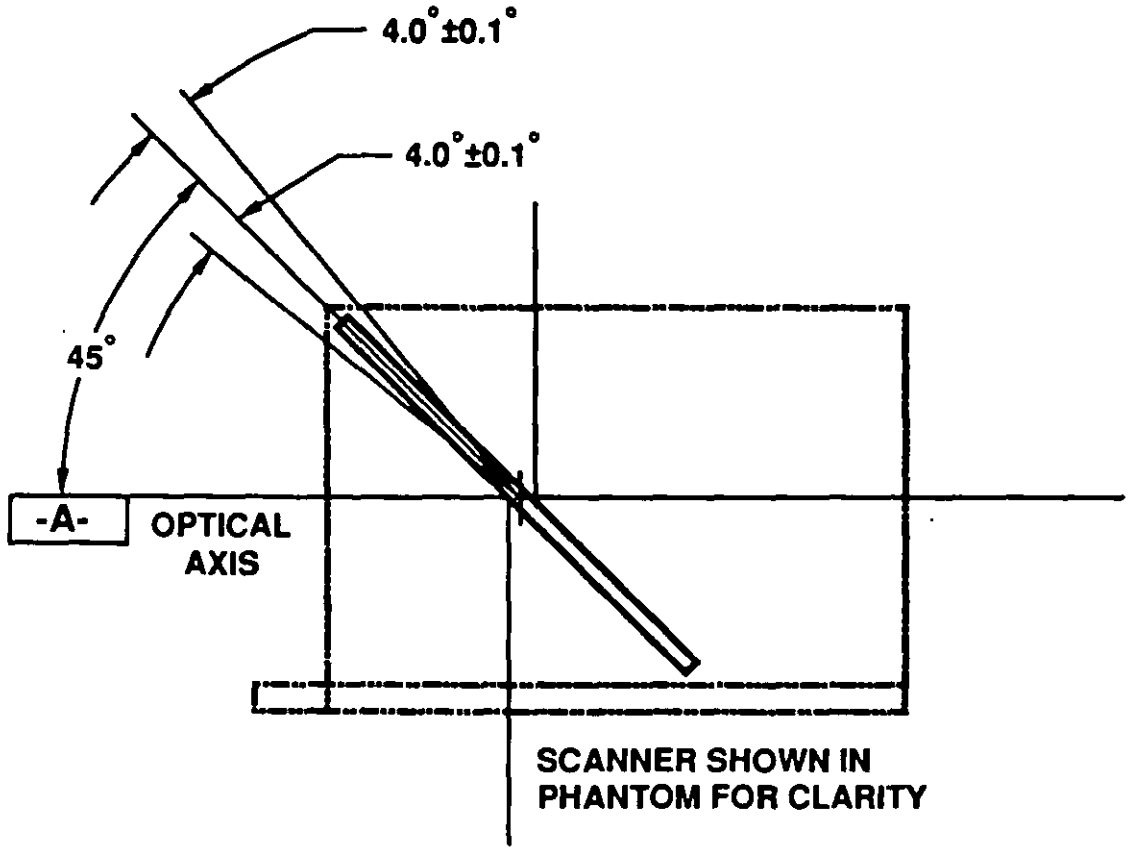
6. NOTES

6.1 Intended use. The MX-9872(V)7/UA is intended for use with the Telescopic Sight Unit of the M-65 C-NITE Thermal Imaging System.

6.2 Other data. See MIL-S-49166.

6.3 Subject Term (Keyword) Listing.

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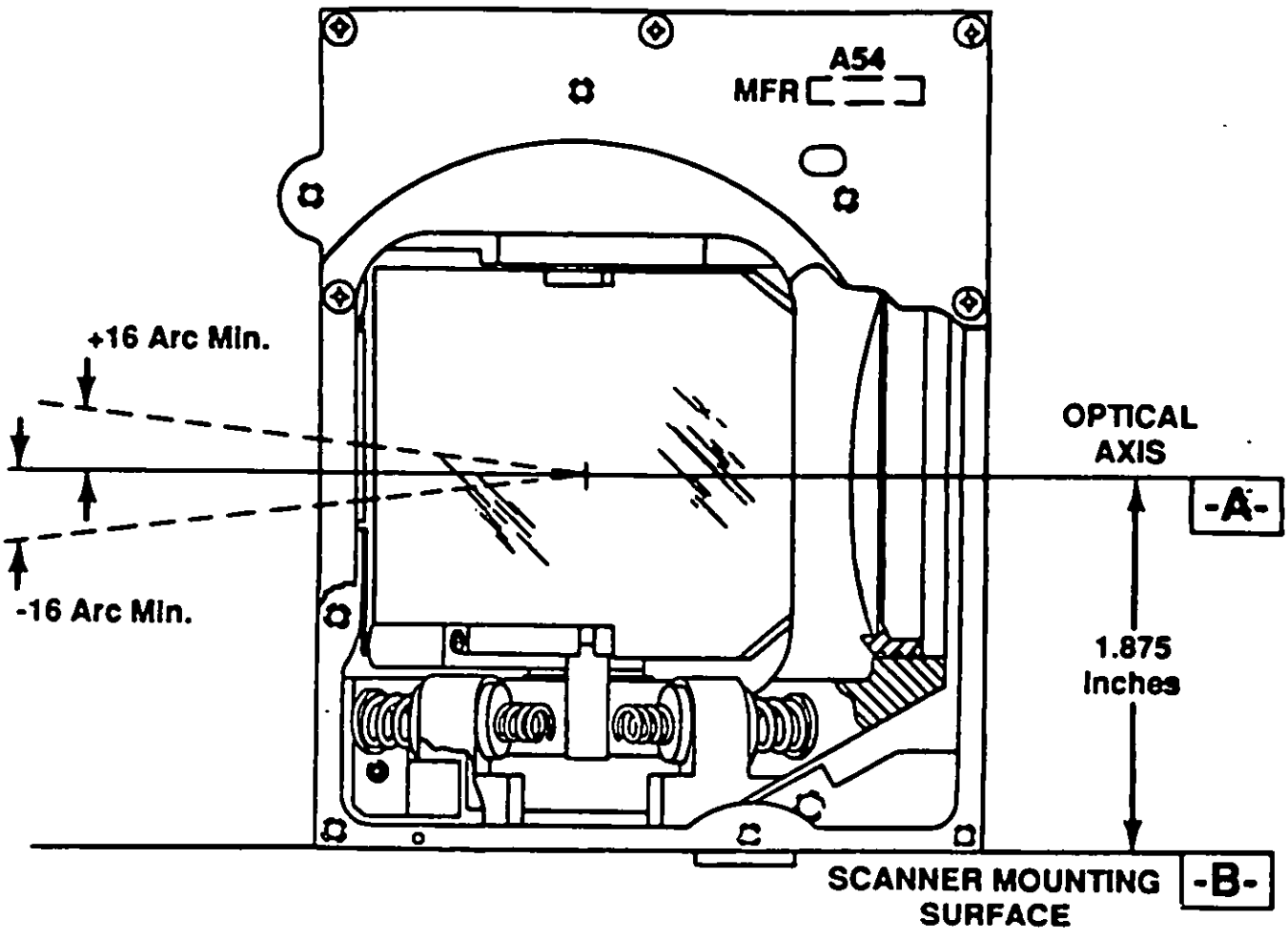


DATUM -A- THE OPTICAL AXIS, IS DEFINED TO BE PARALLEL TO THE MOUNTING SURFACE OF THE SCANNER.

THE AZIMUTH SCAN ANGLE IS DYNAMICALLY SET FOR THE TOTAL SCAN ANGLE OF 8.0°±0.2°. THE SCAN ANGLE IS TO BE +4.0°±0.1° AND -4.0°±0.1° FROM THE MIRROR POSITION ORIENTED AT 45° FROM THE SCANNER DATUM -A-

FIGURE 1 SCAN ANGLE

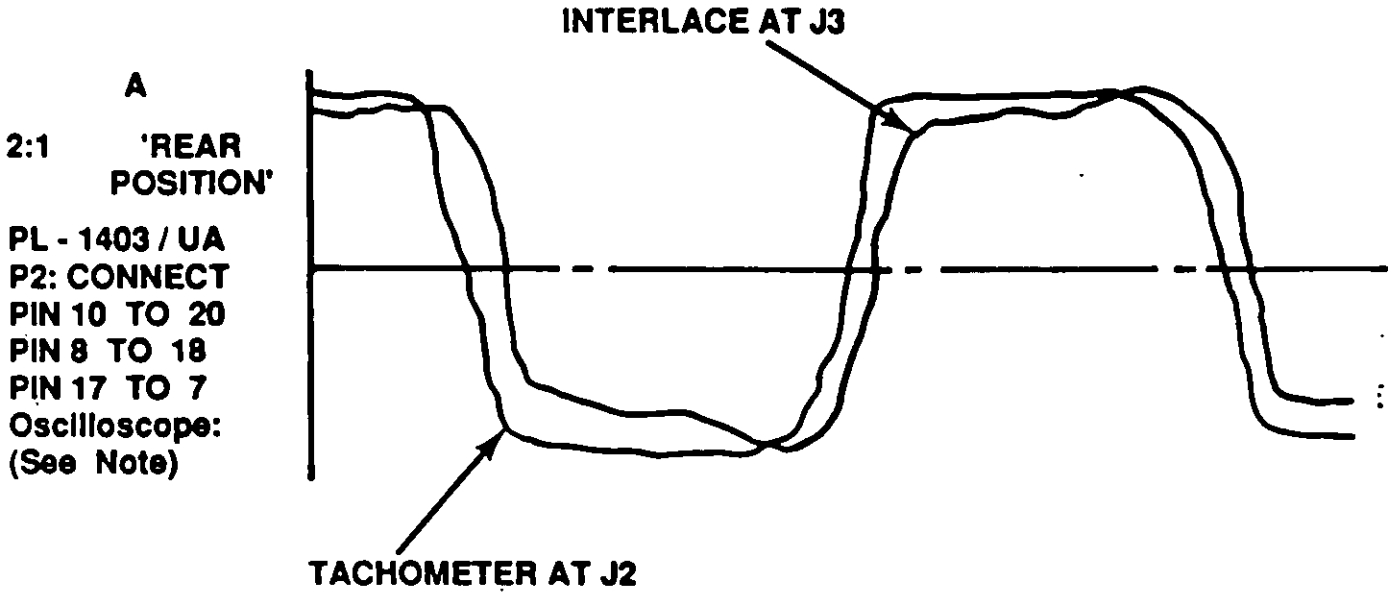
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NOTE: THE OPTICAL AXIS (DATUM -A-) IS PARALLEL TO THE MOUNTING SURFACE, DATUM -B-.

FIGURE 2 ELEVATION BORESIGHT

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NOTE - CONNECT OSCILLOSCOPE AS FOLLOWS:

CHANNEL 1 - (TACHOMETER) HIGH INPUT TO J2 (OR J1 - 13) AND LOW SCOPE TO GROUND.

CHANNEL 2 - (INTERLACE) HIGH INPUT TO J3 (OR J1 - 1) AND LOW SCOPE TO GROUND.

FIGURE 3 **SHAPE AND PHASE RELATIONSHIP OF INTERLACE TRANSDUCER AND SCAN TACHOMETER.**

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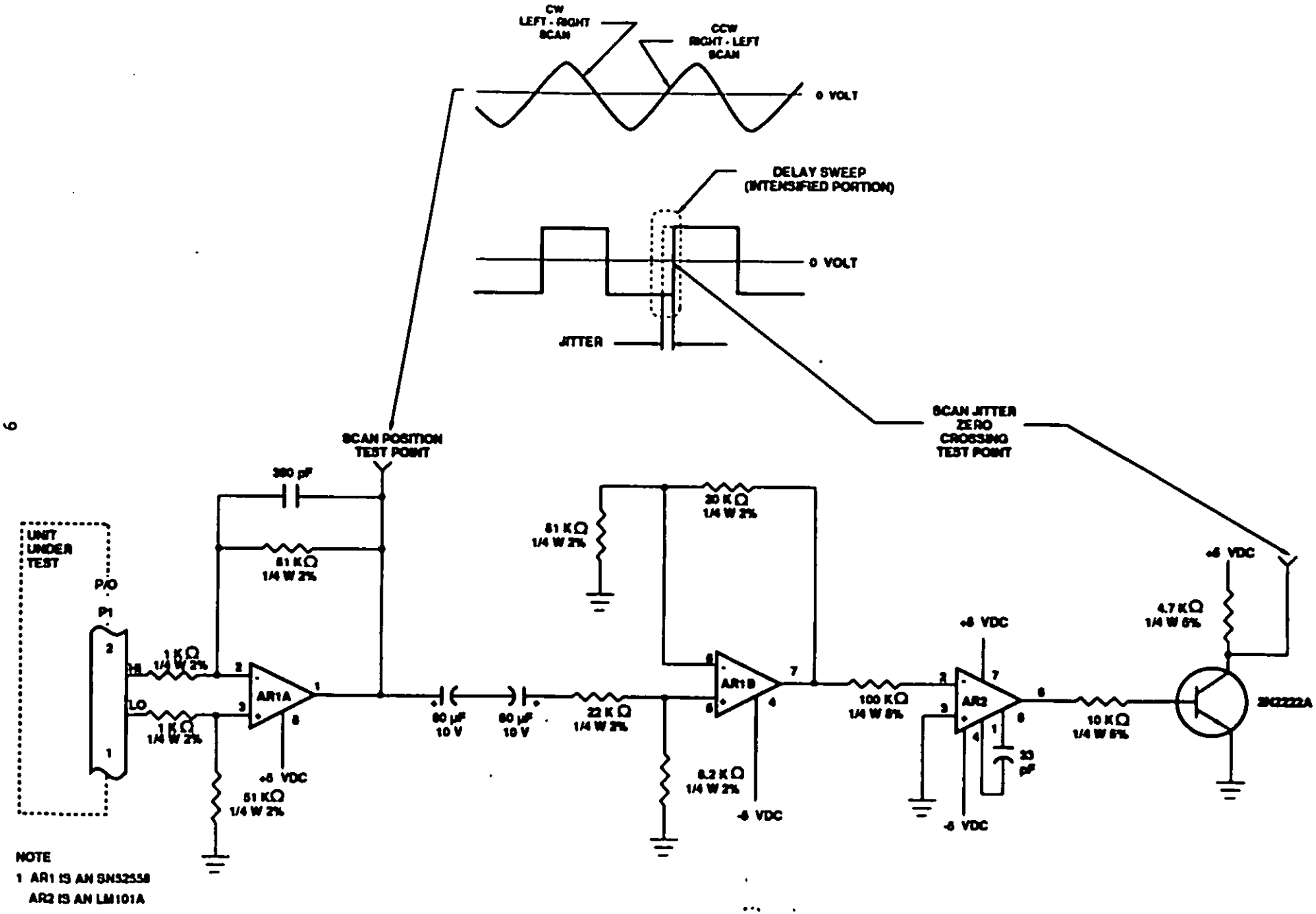


FIGURE 4 TYPICAL CIRCUITRY FOR SCAN JITTER TEST

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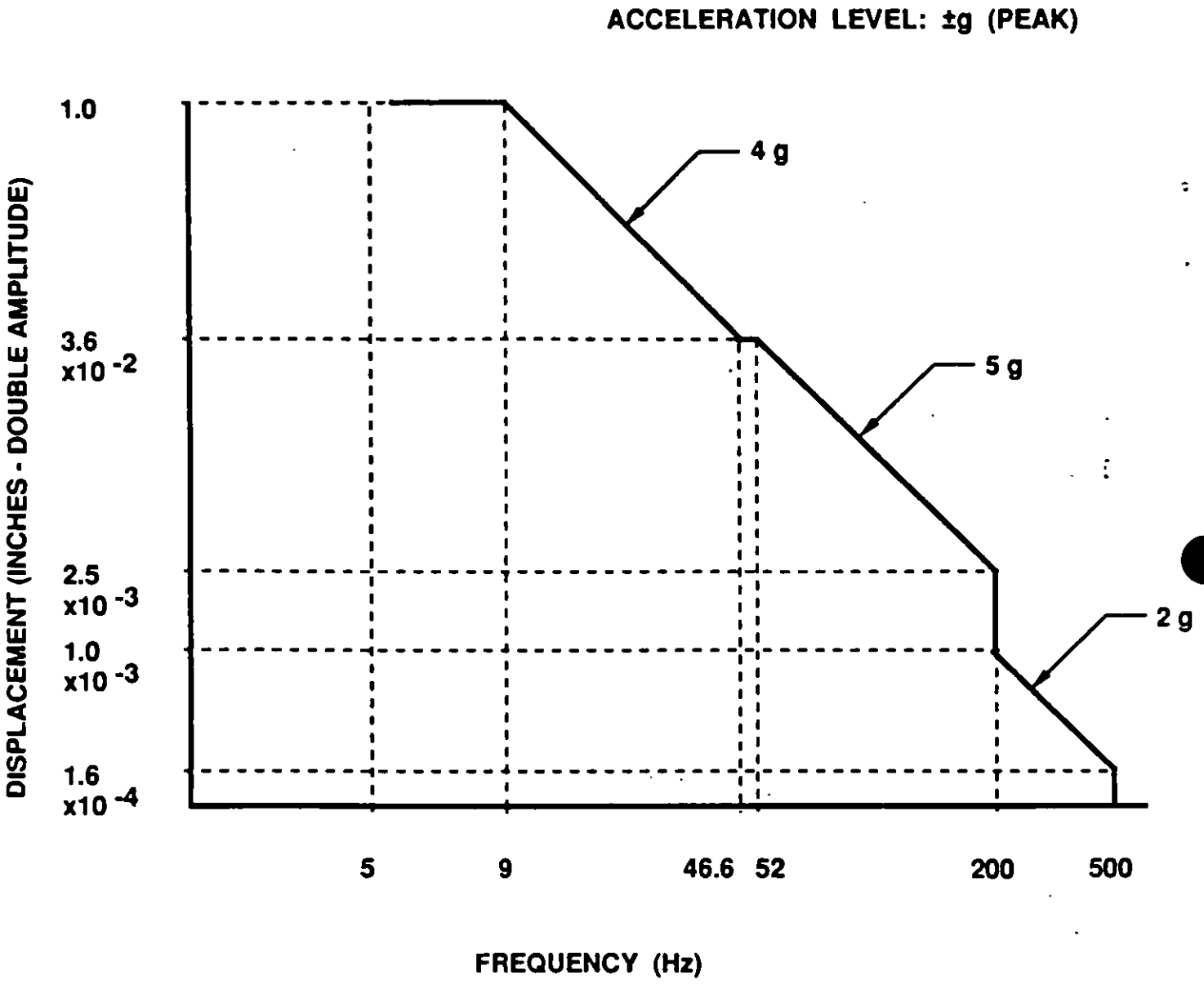


FIGURE 5 VIBRATION TEST PROFILE (X AXIS)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-49166/7(CR)	Scanner, Mechanical MX-9872(V)7/UA

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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